



Fig. 2: IBM's Deep Blue chess machine, which defeated world chess champion Garry Kasparov (Image courtesy: www.scaruffi.com)

ing to achieve results that are not strictly defined.

For example, if we had to create an AI thermostat to cool a house, the program would need to have knowledge of all seasons, weather conditions like El Niño and passage of time, plus it must be able to understand concepts like warm, cool or too cold, apart from other aspects.

While we do not really realise it, the simplest human functions translate to thousands of lines of computer code. Most current AI systems are designed for only a few specific applications. One of the most popular examples of an AI application was a chess program running on Deep Blue, IBM's massively-parallel-computing system.

Deep Blue managed to defeat world chess champion Gary Kasparov because it could search 50 to 100 billion positions in the three minutes that each player had, to make their move.

AI applications can be bucketed as knowledge based or expert systems. A minor knowledge based system could be a series of conditional statements, such as:

IF
the animal is a bird
it does not fly
it swims
it is black and white
THEN it is a penguin.

As this system becomes more complex, the time it takes for a computer to arrive at an intelligent outcome becomes unacceptably high. Expert systems try to solve this issue by acquiring more knowledge from a human being by asking questions. Over time, the program learns from

experience and can actually solve problems or give advice based on what it has learned.

Some application areas of AI

AI is being used in all spheres of everyday life in developed countries. Given below are some key areas.

Online shopping. e-tailing sites such as www.amazon.com deploy a process called collaborative filtering to compare a customer's purchase patterns with those of other customers and provide suggestions. AI can take such processes to the next level to increase loyalty.

In the near future, we can expect semi-autonomous agents surfing the Web to help us, that is, their creators with diverse tasks. Intelligent bots will shop for you online, do your financial transactions and more, without your intervention. You would only be required to provide the bots broad inputs with respect to what you want to get done. Websites such as www.AskJeeves.com are using AI in the hope of making the Internet a more intuitive place, where you get things done the same way you would do in the real world.

e-broking. In the Internet trading environment, an e-broker is a system where clients are the prospective buyers and sellers. Each client has a particular wish-list. In security trading, a buyer can specify type of product, number of lots, maximum purchase price, expiry date, time and other details.

The seller can specify product, number of lots, minimum selling price, expiry date, time and the like. The system preserves a database of outstanding requests from prospective buyers and sellers.

For a buyer, the system screens and shortlists a reasonable number of sellers for further analysis. The system matches buyers and sellers in the shortlist. For each buyer-seller pair, the system makes use of a set of rules and comes up with a

rating for the pair. It ranks possible sellers according to the ratings. Top-ranking sellers are recommended to the buyer.

In such a brokerage system, AI techniques are applied to the short-listing stage and the matching stage. The amount of computation increases sharply with a rise in the number of clients and requests. The focus of research is therefore on efficient strategies and algorithms so that the system can respond to clients' requests within a short span of time. A security trading system is expected to complete several thousand transactions every day and the system has to answer in just a few seconds.

Bioinformatics. AI is critical for the evolution of bioinformatics. Presently, molecular biologists are involved in some notable data-collection projects. Latest genome-sequencing projects are producing a huge volume of data linked to the function and structure of biological molecules and sequences.

Other complementary high-throughput technologies, such as DNA micro-arrays, are swiftly generating big amounts of data that are too overpowering for traditional methods for biological data analysis.

Understanding of this rich data could deeply impact our interpretation of life at the molecular level. However, the illustration of biological knowledge is a very daunting job and increasingly demands more potent and refined computational tools.

AI and other heuristic methods (in particular, machine learning, data mining, cluster analysis, pattern recognition and knowledge representation) could possibly offer key solutions for the fresh hurdles posed by the progressive transformation of biology into a data-massive science.

Some key areas where AI approaches are specifically encouraging and turning out to be fruitful are for prediction of protein's structure and function, semiautomatic drug design, interpretation of nucleotide